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A compact space having the cardinality of the continuum with no convergent sequences

BY V. V. FEDORČUK

Mechanics and Mathematical Faculty, Moscow University, USSR

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1. *Introduction.* All spaces in this paper are Hausdorff. We recall that a space X is sequentially compact, if every countable subset of X contains a convergent sequence. Let us consider the three statements:

- (1) Every compact space of cardinality $\leq c$ contains a point of countable character.
- (2) Every compact space of cardinality $\leq c$ is sequentially compact.
- (3) Every infinite compact space of cardinality $\leq c$ contains a convergent sequence.

Čech and Pospíšil (3) proved that if $\chi(x, X) \geq \tau$ for every point x of a compact space X , then $|X| \geq 2^\tau$. So the Continuum Hypothesis implies that statement 1 is true. On the other hand, if $2^\omega = 2^{\omega_1}$ then the Tychonoff cube I^{ω_1} shows that statement 1 is not true. So statement 1 is independent of ZFC.

Malychin and Šapírovski (7), using results of Booth (2), proved that Martin's Axiom (see for example (6)) implies that statement 2 and so statement 3 are true.

In the present paper we formulate the Partition Hypothesis (PH), which is compatible with ZFC, and show that PH implies the existence of a compact space of cardinality c , which has no isolated points and no convergent sequences. So statements 2 and 3 are independent of ZFC.

2. *Partition hypothesis.* Let $\mathcal{M} = \{M\}$ be a family of subsets of ω . We shall say that $\mathcal{M}$ is a full family of partitions on ω , if for every infinite set $N \subset \omega$ there is a set $M \in \mathcal{M}$ such that both $N \cap M$ and $N \setminus M$ are infinite.

PH. $2^\omega = 2^{\omega_1}$ and there exists a full family of partitions on ω with cardinality ω_1 .

THEOREM 1. $\text{Con ZFC} \Rightarrow \text{Con (ZFC + PH)}$.

The proof is analogous to the proof of the main theorem in (5). Here is an outline: suppose the universe is the result of adding a sequence $\langle X_\nu : \nu < \omega_2 \rangle$ of ω_2 Cohen generic

subsets of ω to the constructible universe L . Then $2^\omega = 2^{\omega_1} = \omega_2$. Let $\mathcal{M}$ be $\{X_\nu: \nu < \omega_1\}$. Given $N \subseteq \omega$, there is a countable subset Z of ω_2 such that $N \in L[\langle X_\nu: \nu \in Z \rangle]$, as from the point of view of L , N depends on only countably many co-ordinates of the product space $2^{\omega \times \omega_2}$; so if $\mu \in \omega_1 \setminus Z$, X_μ is Cohen generic over $L[N]$, and hence if N is infinite both $N \cap X_\mu$ and $N \setminus X_\mu$ will be infinite.

3. *Preliminary construction and results.* Let X be a topological space and let $a = (F_a, Y_a, f_a)$, where F_a is a closed subset of X , Y_a is a topological space and

$$f_a: X \setminus F_a \rightarrow Y_a$$

is a map.

We put $K(X, a) = F_a \times Y_a \cup \text{Graph } f_a$. The set $K(X, a)$ is closed in $X \times Y_a$. If $F_a = X$, we shall assume that $K(X, a) = X \times Y_a$. We shall denote by $\pi_a: K(X, a) \rightarrow X$ the restriction of the projection $X \times Y_a \rightarrow X$ to $K(X, a)$.

Now let $\mathfrak{A} = \{a\}$ be a family of the triples $a = (F_a, Y_a, f_a)$. We denote by $\pi^{\mathfrak{A}}: X^{\mathfrak{A}} \rightarrow X$ the fibre-product of the family $\{\pi_a: K(X, a) \rightarrow X\}_{a \in \mathfrak{A}}$, that is

$$X^{\mathfrak{A}} = \{(z_a) \in \prod_{a \in \mathfrak{A}} K(X, a): \pi_a z_a = \pi_{a'} z_{a'} \text{ for all } a, a' \in \mathfrak{A}\}$$

and $\pi^{\mathfrak{A}} = \pi_a p_a$ for all $a \in \mathfrak{A}$, where $p_a: \Pi\{K(X, a'): a' \in \mathfrak{A}\} \rightarrow K(X, a)$ is the natural projection.

The construction of $X^{\mathfrak{A}}$ and $\pi^{\mathfrak{A}}$ is a slightly different version of that given by Ulyanov (8). If each F_a consists of one point, then this construction coincides with the one given by the author in (4).

If X and all Y_a are compact, then $X^{\mathfrak{A}}$ is compact, being a closed subset of the compact space $\Pi K(X, a)$. Further, for every point $x \in X$ the inverse image $(\pi^{\mathfrak{A}})^{-1}x$ is homeomorphic to $\Pi\{Y_a: a \in \mathfrak{A}, F_a \ni x\}$. So we have

LEMMA 1 (see (7)). *Let X and all Y_a be first countable compact spaces and let $\{F_a: a \in \mathfrak{A}\}$ be a point-countable family. Then $X^{\mathfrak{A}}$ is also a first countable compact space.*

LEMMA 2. *Let X and all Y_a be zero-dimensional compact spaces. Then $\dim X^{\mathfrak{A}} = 0$.* This follows, for example, from the Hurewicz formula

$$\dim X^{\mathfrak{A}} \leq \dim X + \dim \pi^{\mathfrak{A}}.$$

4. Killing of sequences.

Definition 1. Let $C \subset X$ be a convergent sequence, that is C is a countable discrete set such that $\bar{C}$ is compact and $\bar{C} \setminus C$ consists of one point. We shall say that a map $f: Y \rightarrow X$ kills the sequence C if there is no convergent sequence $B \subset Y$ such that $fB = C$.

LEMMA 3. *Let $f: Y \rightarrow X$ and $g: Z \rightarrow Y$ be maps. If f kills a convergent sequence $C \subset X$, then fg kills C as well.*

The proof is trivial.

Definition 2. We shall say that an inverse spectrum $S = \{X_\alpha, \pi_\alpha^{\alpha'}: \alpha, \alpha' \in A\}$ kills all convergent sequences, if for every $\alpha \in A$ and for every convergent sequence $C \subset X_\alpha$ there is an $\alpha' \in A$ such that $\alpha' > \alpha$ and $\pi_\alpha^{\alpha'}$ kills C .

LEMMA 4. If a well-ordered spectrum $S = \{X_\alpha, \pi_\alpha^{\alpha'}\}$ of length ω_1 kills all sequences, then $\lim S$ has no convergent sequences.

Proof. We suppose that there is a convergent sequence $C \subset \lim S$. Let $\bar{C} \setminus C = \{x\}$. There is an $\alpha < \omega_1$ such that $\pi_\alpha C$ is infinite and so $D = \pi_\alpha C \setminus \{\pi_\alpha x\}$ is a convergent sequence. Then $E = C \setminus \pi_\alpha^{-1} \pi_\alpha x$ is a convergent sequence, being an infinite subset of a convergent sequence. Moreover $\pi_\alpha E = D$. But this contradicts Lemma 3 and the killing property of S .

Let $\mathcal{M} = \{M_\alpha: \alpha < \omega_1\}$ be a full family of partitions on ω .

For every first countable zero-dimensional compact space X and every point $x \in X$ we fix a nested base $\{O_m x: m \in \omega\}$ of open-and-closed neighbourhoods of x with $O_0 x = X$.

Now for every $\alpha < \omega_1$ we define the map

$$f_{(x, X)}^\alpha: X \setminus \{x\} \rightarrow D = \{0, 1\}$$

in the following way:

$$(f_{(x, X)}^\alpha)^{-1}(0) = \cup \{O_m x \setminus O_{m+1} x: m \in M_\alpha\}.$$

Definition 3. Let X be a zero-dimensional first countable compact space and let $C \subset X$ be a sequence, converging to the point $x \in X$. We shall say that an ordinal α is the index of the sequence C , if $\alpha = \min \{\beta: \text{both } N \cap M_\beta \text{ and } N \setminus M_\beta \text{ are infinite}\}$, where $N = \{m \in \omega: C \cap (O_m x \setminus O_{m+1} x) \neq \emptyset\}$.

LEMMA 5. Let $g: X \rightarrow Y$ be a map from a compact space X onto a zero-dimensional first countable compact space Y and let $C \rightarrow Y$ be a sequence of index α , converging to a point $y \in Y$. Then the map $g\pi_\alpha: K(X, a) \rightarrow Y$, where $a = (g^{-1}y, D, f_{(y, Y)}^\alpha, g)$, kills the sequence C .

Proof. We assume that there is a sequence $B \subset K(X, a)$, converging to some point $z \in \pi_\alpha^{-1}g^{-1}y$, such that $g\pi_\alpha B = C$. We put

$$B_0 = B \cap (\cup \{\pi_\alpha^{-1}g^{-1}(O_m y \setminus O_{m+1} y): m \in M_\alpha\})$$

and $B_1 = B \setminus B_0$. Both B_0 and B_1 are infinite, because the index of C is α . So

$$(\pi_\alpha z, i) \in \bar{B}_i,$$

by the definition of $K(X, a)$. Hence $|\bar{B} \setminus B| \geq 2$. This contradicts the convergence of B . Lemma 5 is proved.

5. Dyadic resolution. Let $S_\alpha = \{X_\beta, \pi_\beta^{\beta'}: \beta < \alpha\}$ be a continuous countable spectrum, whose objects are zero-dimensional first countable compact spaces. For a limit ordinal α we put $X_\alpha = \lim S_\alpha$ and $\pi_\alpha^\beta = \lim_{\beta' < \alpha} \pi_\beta^{\beta'}$. If $\alpha = (\alpha - 1) + 1$, then $X_\alpha = X_{\alpha-1}^{\mathfrak{Q}_\alpha}$, where

$$\mathfrak{Q}_\alpha = \{((\pi_\beta^{\alpha-1})^{-1}x, D, f_{(x, X_\beta)}^\gamma, \pi_\beta^{\alpha-1}): x \in X_\beta, \beta, \gamma < \alpha\},$$

and

$$\pi_{\alpha-1}^\alpha = \pi^{\mathfrak{Q}_\alpha}, \pi_\beta^\alpha = \pi_\beta^{\alpha-1} \pi_{\alpha-1}^\alpha.$$

The spectrum $S_{\alpha+1}$, which is the union $S_\alpha \cup \{X_\alpha, \pi_\beta^\alpha: \beta \leq \alpha\}$, will be called *the dyadic α -extension of S_α with respect to $\mathcal{M}$* (the maps $f_{(x, X_\beta)}^\gamma$ depend on $\mathcal{M}$).

LEMMA 6. *Let X be an arbitrary zero-dimensional first countable compact space and let $1 \leq \beta \leq \omega_1$. Then there exists a unique continuous spectrum*

$$S_\beta = S_\beta(X) = \{X_\alpha, \pi_\alpha^{\alpha'}: \alpha, \alpha' < \beta\}$$

of length β , whose objects are zero-dimensional first countable compact spaces, $X_0 = X$ and $S|\alpha + 1$ is the dyadic α -extension of $S|\alpha$ for each $\alpha < \beta$.

Proof. Uniqueness is trivial. Existence is by induction. Let $S_1 = \{X = X_0, 1_X\}$. For a limit ordinal α we put $S_\alpha = \cup\{S_{\alpha'}: \alpha' < \alpha\}$. If $\alpha = (\alpha - 1) + 1$, then S_α is the dyadic $(\alpha - 1)$ -extension of $S_{\alpha-1}$. One needs only to check that $X_{\alpha-1}$ is a zero-dimensional first countable compact space. If $\alpha - 1$ is a limit ordinal, then $X_{\alpha-1}$ is a zero-dimensional first countable compact space, being a limit of a countable spectrum $S_{\alpha-1}$, whose objects are zero-dimensional first countable compact spaces. If $\alpha - 1 = (\alpha - 2) + 1$ then $X_{\alpha-1} = X_{\alpha-2}^{\omega_1}$ and so $X_{\alpha-1}$ is a zero-dimensional first countable compact space by Lemmas 1 and 2. Lemma 6 is proved.

Definition 4. The spectrum $S_\beta(X)$, whose existence was proved in Lemma 6, will be called *the dyadic β -resolution of the space X with respect to $\mathcal{M}$* .

6. Counterexample.

LEMMA 7 (PH). *Let X be an arbitrary zero-dimensional first countable compact space and let $D(X) = \lim S_{\omega_1}(X)$, where $S_{\omega_1}(X)$ is the dyadic ω_1 -resolution of X with respect to $\mathcal{M}$. Then $D(X)$ is a compact space of cardinality $\mathfrak{c}$ with no convergent sequences and no isolated points.*

Proof. By Archangelskii's theorem (1), $|X_\alpha| \leq \mathfrak{c}$ for all $X_\alpha \in S_{\omega_1}(X)$. So

$$|D(X)| \leq \mathfrak{c}^{\omega_1} = \mathfrak{c}.$$

Further, for every $\alpha < \omega_1$ each point $x \in X_\alpha$ has non-trivial inverse image under $\pi_\alpha^{\alpha+1}$. Therefore $|X_\alpha| \geq \mathfrak{c}$ for all α such that $\omega \leq \alpha < \omega_1$. Hence $|D(X)| = \mathfrak{c}$. On the other hand, the property of $\pi_\alpha^{\alpha+1}$, just mentioned, implies that for every limit ordinal α the space X_α has no isolated points (in fact it is true for all $\alpha \geq \omega$). Hence $D(X)$ has no isolated points either.

Finally Lemmas 3 and 5 imply that the spectrum $S_{\omega_1}(X)$ kills all sequences, which have countable indices. But every sequence has a countable index, because $\mathcal{M}$ is a full family of partitions on ω with cardinality ω_1 . So according to Lemma 4 the space $D(X)$ has no convergent sequences. Lemma 7 is proved.

From Theorem 1 and Lemmas 6 and 7 we get:

THEOREM 2. $\text{Con ZFC} \Rightarrow \text{Con (ZFC + there exists a compact space of cardinality } \mathfrak{c}, \text{ which has no convergent sequences and no isolated points)}.$

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